

List of Prompts

1.give me the system architecture for

Track 1: KOHLER AI Bathroom Designer & Planner

Objective: Build an interactive AI design assistant that takes a customer's constraints and automates personalized product bundle recommendations.

Key Inputs: Bathroom dimensions (ft x ft / layout or image), budget limits, aesthetic themes (e.g., Minimalist Modern, Classic Luxury, Japanese Zen), and device catalog specifications.

Expected Outcome: An intelligent recommendation engine that outputs optimized product combinations (faucets, smart toilets, thermostatic showers, vanities) fitting exact physical space and budget parameters. If possible a 2D or 3D representation of the bathroom layout with the selected products.

2.Can you help me design the system prompts and JSON schemas for the AI Orchestrator layer to extract dimensions and constraints?

3.Give me the AI prompts for each layer and guide me which LLM will work the best on it.

4.remember that all the apis and other features should be open source and free.

5.Can you guide me on how to set up Ollama and ChromaDB locally for this project using Python?

6.tell me how to implement frontend.

7.could you give me appropriate info for using codex and attaching appropriate plugins for an amazing 3d website

8. Understand the existing codebase

Uploaded the existing project (index.html, app.js, styles.css, PROMPTS.md, README.md, setup_db.py,

query_db.py) with the request: "read these file and understande my concept first."

9. Architecture review + first feature request

Uploaded a strategy document (an AI-generated "Track Selection Strategy" PDF covering system architecture

recommendations) with the request: "read the architecture from this pdf and add some interesting features

according to you like clicking a photo and taking idea of the place.” This led to the Photo Style Match feature

(on-device color analysis of an uploaded bathroom photo).

10. Site not loading on GitHub Pages

“give me the code to be pasted in index.html file as the prototype is not opening using the public url provided by github.”

11. Live URL diagnosis

Shared the live GitHub Pages URL directly for inspection.

12. Cloudflare Pages/Workers confusion

“should I connect github to workers and pages...it is asking for application creation.”

13. Deployment mechanics

“so i have to paste the new codes as well in my file??”

14. Git merge conflict resolution

A multi-turn debugging exchange: pasted a `git push` rejection error, then raw merge-conflict markers from a

failed pull, then the resulting broken/duplicated HTML file — with the recurring question “so i shoul accept the

changes??”, “any problem??”, and “open the file and check yourself once.”

15. Bug report

“the ai moodboard is still jittery...cannot enter data into it.” (Root cause found later: a CSS specificity bug

overriding the HTML hidden attribute.)

16. UI walkthrough request

17. Data integrity question

“How many dataset from kohler have been used.”

18. Data realism upgrade

“We have to add datasets” — clarified via a follow-up choice as: “More realistic data — real Kohler names/specs instead of invented numbers.” This led to replacing the invented product catalog with real, sourced Kohler product

data (documented in DATA_SOURCES.md).

19. Free API/LLM integration ideation

“what changes can be done for good if I integrate APIs and other LLMs (free of course) for the project.”

20. Implementation request

“do it and also tell me how to add them.” This added the Designer's Note (AI-written bundle pitch) and an AI Moodboard image feature.

21. Secure backend request

“you know what...lets make backend but in a safe way.” This led to a Cloudflare Worker proxying the Designer's Note through Groq, with the API key held server-side only.

22. Backend deployment walkthrough

A guided, multi-turn deployment: “I created the key...now what,” “should I connect github to workers and pages...,”

“so i have to paste the new codes as well in my file??”

23. Large feature-set request

Uploaded a 14-item feature brainstorm document (design tiers, trade-off mode, product swap, interactive 2D planner, plumbing awareness, conversational loop, version history, etc.) with the request: “add these features as well if possible.”

24. Scope triage decision

“do the features that can be done right now and avoid the ones too risky.” This resulted in shipping: design tiers (Essential/Signature/Sustainable), a Why-This-Works checklist with confidence breakdown, trade-off diagnosis for infeasible constraints, per-product swap with real alternatives, a sustainability progress bar, and Before/After photo framing — while explicitly skipping the interactive drag-and-drop planner, plumbing awareness, and full multimodal orchestration as too high-risk for the remaining time.

25. Deployment clarification

“which files to edit where” and “but the backend enhances it right?”

26. Live preview failure diagnosis

“why is the live preview not available.”

27. Root-cause evidence

Pasted a live 402 “Insufficient Balance” error from the Pollinations image API, confirming the free image-generation tier had moved to a paid credit system.

28. Follow-up visual gap report

“i still cant see the image” — revealing that the interim fallback (a static decorative card) was not actually a photograph, prompting a proper zero-dependency SVG illustration fallback.

29. Real photography request

“I need an image of a designated bathroom there according to the description given by the image.” This led to integrating the Unsplash Search API (licensed, attributed real photography) as the primary source, with the SVG illustration as an automatic no-key fallback.

30. Finalization

“continue....give me the final files, no upgrading now and no errors as well.”

WORKFLOWS

